
ORF 245 FUNDAMENTALS OF STATISTICS

Spring 2025

Last Updated: February 5, 2025

Instructor:

- Dan Rigobon: drigobon@princeton.edu
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Head TA:

- Jiawei Ge: jg5300@princeton.edu
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Teaching Assistants:

- Silu Li: sl3965@princeton.edu
 - Dimitrios Lolas: dl7200@princeton.edu
 - Silvia Tao: zhixu.tao@princeton.edu
 - Giang Truong: gt6503@princeton.edu
 - Chang Yu: cy7438@princeton.edu
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Lectures:

Weekly, in-person, held on: Mondays & Wednesdays from 3:00 - 4:20 pm in CS 104

Weekly Precepts (all covering the same material):

Attendance at precepts is not mandatory, but is strongly encouraged. Precepts will focus on: i) review of concepts from lectures, ii) examples and practice problems, and iii) working more closely with R.

- Section P01: Monday, 7:30 pm - 8:20 pm
TA: Chang Yu, Room: Friend 108
- Section P02: Monday, 7:30 pm - 8:20 pm
TA: Dimitrios Lolas, Room: Friend 109
- Section P03: Tuesday, 3:30 pm - 4:20 pm
TA: Giang Truong, Room: Friend 108
- Section P04: Tuesday, 3:30 pm - 4:20 pm
TA: Silu Li, Room: Friend 109
- Section P05: Tuesday, 7:30 pm - 8:20 pm
TA: Silvia Tao, Room: Friend 108
- Section P06: Tuesday, 7:30 pm - 8:20 pm
TA: Chang Yu, Room: Friend 109

Instructor Office Hours:

- Dan Rigobon: Wednesdays, 11am-12pm, Sherrerd 204

Purpose: Primarily for questions about course material and about concepts from class. (Note: Questions about homework should be primarily addressed to TA's during precepts and office hours. For any course logistics and grading inquiries (e.g.'s: re-grading, homework extensions) email the course Head TA, cc'ing Daniel Rigobon.)

TA Office Hours/Help Sessions:

- Mondays, 7:30-8:30pm, Sherrerd 123
- Tuesdays, 2-3pm, Sherrerd 003
- Wednesdays, 2-3pm, Sherrerd 107
- Wednesdays, 7:30-8:30pm, Sherrerd 107
- Thursdays, 2-3pm, Sherrerd 107
- Thursdays, 7:30-8:30pm, Sherrerd 107

Purpose: Primarily for questions about homework, although you are welcome to ask about course material or concepts in class. (Note: These hours may be staffed by a different set of TA's every week.)

UCA Office Hours/Help Sessions:

We have three Undergraduate Course Assistants this fall, who will be holding weekly office hours. As with the TA's Office Hours, these are primarily intended to be for homework-related questions, but you may ask anything you like.

- Sofia Arora: sa9267@princeton.edu
Hours: Fridays, 10am-11am, Sherrerd 107
 - Christian Garcia-De La Jara: christiangarcia@princeton.edu
Hours: Thursdays, 4:30pm-5:30pm, Sherrerd 107
 - Anany Kotawala: ak6295@princeton.edu
Hours: Mondays, 6pm-7pm, Sherrerd 003
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Main Reference:

- **Book:** Jay Devore, Probability and Statistics for Engineering and the Sciences, 9th Edition.

We will not be covering all of the book's material, and so the textbook is intended to be a supplement (but not a substitute) for lectures and precepts.

Note: The 9th edition has gone through some major changes in chapters 8-15, and new exercises have been added in chapters 2-5.

- **Additional resources:** There are many books on probability and statistics at various levels. There is also a lot of material (articles, videos, etc.) online. You are welcome to use any of these resources to help support your learning, and we will occasionally be posting relevant external content on Ed.

Prerequisites:

It is essential to have a good understanding of calculus and some exposure to multivariate calculus. It will also be valuable, but not essential, to have some mathematical maturity (e.g. exposure to proof-based mathematics).

We will not formally teach programming in the language R, but you will be expected to pick it up through precepts, guided exercises, and resources available on campus and online. It is recommended that students download and install both R and RStudio before the first class:

R: <https://cran.r-project.org>

RStudio: <https://posit.co/products/open-source/rstudio/>

Some online tutorials to R can be found here:

<https://www.tutorialspoint.com/r/index.htm>

<https://www.w3schools.com/r/>

Course Components and Grading:

- *Homework/problem sets (30%)*. There will be eight graded problem sets throughout the semester. These will be due on Fridays at end-of-day (11:59pm) on specified weeks, with submission on Gradescope.

We encourage you to work together and discuss homework problems. However, you should write up and submit your solutions independently based on your own understanding of the problems, and without reading or using the solutions of other students or other online resources. We also encourage you to attend precepts and TA office hours if you would like to discuss any of the homework problems.

Late Policy: 25% will be deducted for each day late, of course unless you've received an extension on the assignment.

Regrade Policy: Any requests need to be submitted on Gradescope no more than one week after grades for a homework assignment have been released.

Note that re-grades are to assess whether the rubric (determined by the TA's) has been uniformly applied to students, and not to question the validity of the rubric itself.

- *Midterm exam (30%)*. There will be one in-class midterm exam (of 1 hour and 20 minutes in duration), tentatively in-class from **3pm-4:20pm on Wednesday March 5th**. The exam will be open book and open notes, but you are prohibited from accessing any online or external resources. You will also be allowed to use a calculator or R for performing simple computations.
 - *Final exam (40%)*. There will be a final exam as scheduled by the Registrar during final exam period. The final exam will be comprehensive, covering all material taught during this semester. Like the midterm exam, it will be open book and open note, with no online access.
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AI Policy:

You may use LLM's or other such generative AI tools for: (i) asking questions about concepts in statistics that you may not fully understand, and (ii) assisting with programming in R. Wherever you use these tools for coding purposes, you must cite them appropriately (e.g. in the header of your code file).

Tentative Course Outline:

We will *loosely* keep to the following outline:

1. Introductory and Basic Ideas [Ch. 1] [1 week] [HW 1]
 - Statistics vs. Probability
 - Sample (observations) vs. Population (underlying truth)
 - Descriptive Statistics, Median, Standard Deviation, Variance
2. Probability [Ch. 2] [1 week] [HW 2]
 - Sample Space, Events, Union and Intersection of Events
 - Axioms of Probability
 - Conditional Probability
 - Independence
3. Random Variables and Important Distributions [Ch. 3, Ch. 4, Ch. 5] [3 weeks] [HW 2-4]
 - Discrete and Continuous Random Variables
 - Expected Value
 - Probability Density Function and Cumulative Distribution Function
 - Distributions: Binomial, Hypergeometric, Negative Binomial, Poisson, Normal or Gaussian, Exponential
 - Joint Distributions of Random Variables
 - Weak Law of Large Numbers and Central Limit Theorem
4. Concepts and Methods of Estimation [Ch. 6, Ch. 7] [2 weeks] [HW 5]
 - Estimators, Unbiased Estimation, Minimum Variance Unbiased Estimation
 - Standard Error
 - Maximum Likelihood Estimation
 - Confidence Intervals, Precision, Sample Size
 - Intervals Based on Normal Population
5. Hypothesis Testing [Ch. 8, Ch. 9] [2 weeks] [HW 6]
 - p -values, type-I and type-II errors
 - z-Test, t-Test, Test for Population Proportion
 - Inference Based on Two Samples
 - Two-Sample z-test, Two-Sample t-test
 - Analysis of Paired Data
6. Regression [Ch. 12, Ch. 13] [2 weeks] [HW 7-8]
 - Simple Linear Regression
 - Estimation of Model Parameters
 - Multiple and Nonlinear Regression (polynomial and other)
7. Advance Topics (if time permits) [1 week]